

Divide and Summarize

A slice operation on an array of integers S is defined as follows. If $\max(S) = \min(S)$, the operation cannot be applied. Otherwise, let

$$m := \left\lfloor \frac{\max(S) + \min(S)}{2} \right\rfloor$$

S is partitioned into two arrays: S_1 , containing the elements of S that are less than or equal to m , and S_2 , containing the elements of S greater than m . Then, S is replaced with either S_1 or S_2 . An array is called *reachable* from S if it is the result of some sequence of slicing operations on S .

You are given an array A of length n , and an array **queries** of length q . For each i , determine if there is some array reachable from A whose sum is **queries**[i].

($\lfloor \cdot \rfloor$ is the *floor function*, which rounds any number down to the nearest integer.)

Example Test Cases:

Test Case 1

Input: $n = 5$, $A = [1, 2, 3, 4, 5]$, $q = 5$, **queries** = $[1, 8, 9, 12, 6]$

Output: $[1, 0, 1, 0, 1]$

Explanation: The array A initially has a sum of 15. $m = 3$, so we can replace A with $[4, 5]$, which has a sum of 9. However, no array with a sum of 8 or 12 is reachable.

Test Case 2

Input: $n = 5$, $A = [3, 1, 3, 1, 3]$, $q = 5$, **queries** = $[1, 2, 3, 9, 11]$

Output: $[0, 1, 0, 1, 1]$

Explanation: The array A initially has a sum of 11. $m = 2$, so we can replace A with $[1, 1]$, which has a sum of 2. Because $\max([1, 1]) = \min([1, 1]) = 1$, no further slices can be applied. No array with a sum of 1 or 3 is reachable.

Expected Time Complexity: $O(n \log(\max(A)) + q)$

Input Format:

An integer n , an array A of length n , an integer q , and an array **queries** of length q .

Output Format:

Return an integer array of length q , where the i -th element is 1 if **queries**[i] is the sum of an array reachable from A , and 0 otherwise.

Constraints:

1. $1 \leq n, q \leq 10^5$
2. $1 \leq A[i] \leq 10^6$
3. $1 \leq \text{queries}[i] \leq 10^9$

Instructions:

You have been provided with two template files, containing the `main` and `solve` functions respectively. The `main` function, which has been completed for you, parses the input for each test case. For each test case, it calls the `solve` function, which must compute the answers to all queries.

The `solve` function must compute this in $O(n \log(\max(A)) + q)$ time.

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)

Library Usage

For this lab, you are permitted to use any modules which are part of the standard library of either `C++` or `python`. For `C++`, do not add any extra `#include` directives; the provided `bits/stdc++.h` covers all allowed modules.